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EXECUTIVE SUMMARY

This study develops a predictive analytics framework to identify individuals at risk of persistent high healthcare expenditure using longitudinal data from the Medical Expenditure Panel Survey (MEPS). Persistent high-cost individuals were defined as those within the top 10% of total healthcare spending over a two-year period. A survey-weighted logistic regression model was implemented to account for the complex sampling design of MEPS data.

The dataset included multiple panels, with a panel-based temporal split used for validation to improve external validity. Model performance was strong, achieving an AUC of 0.934, indicating excellent discrimination between high-cost and non-high-cost individuals. At a classification threshold aligned with outcome prevalence (10%), the model demonstrated high sensitivity (90.1%), ensuring most high-risk individuals were correctly identified.

Risk stratification analysis revealed substantial separation, with over 50% of individuals in the highest risk quintile classified as persistent high-cost, compared to less than 1% in the lowest quintile. Key predictors included prior healthcare expenditure and chronic conditions such as hypertension, stroke, heart disease, diabetes, and asthma.

The findings highlight the model's potential for practical healthcare applications, including early identification of high-risk populations, targeted interventions, and improved resource allocation.

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INTRODUCTION

Healthcare expenditure in the United States is highly concentrated, with a small proportion of individuals accounting for a significant share of total spending (Cohen & Yu, 2012; Newhouse, 1992). Identifying individuals who are likely to remain high-cost over time is critical for improving healthcare planning, reducing unnecessary costs, and enabling targeted interventions. Persistent high-cost patients often require continuous medical attention due to chronic conditions, comorbidities, and healthcare utilization patterns (Hwang et al., 2001). Early identification of such individuals allows healthcare providers and policymakers to implement preventive strategies and optimize resource allocation.

This study aims to develop a predictive model using MEPS longitudinal data to identify individuals at risk of persistent high healthcare expenditure. The focus is on building an interpretable and statistically robust model that aligns with real-world healthcare decision-making.

LITERATURE REVIEW

Healthcare cost concentration has been widely studied, with evidence showing that a small percentage of individuals account for a disproportionately large share of healthcare expenditures (Cohen & Yu, 2012). Prior research has demonstrated that past healthcare spending is one of the strongest predictors of future costs (Johnson et al., 2015).

Studies using MEPS data have highlighted the importance of chronic conditions such as diabetes, hypertension, and cardiovascular diseases in driving healthcare expenditures (Hwang et al., 2001). Additionally, socio-demographic factors such as income, insurance status, and race have been shown to influence healthcare utilization and cost patterns (Diehr et al., 1999).

Predictive modeling approaches, including logistic regression and machine learning techniques, have been applied to identify high-cost patients (Zhu et al., 2020). Logistic regression remains widely used due to its interpretability and ability to provide clinically meaningful insights through odds ratios.

Survey-weighted modeling is particularly important when working with MEPS data due to its complex sampling design. Incorporating survey weights ensures unbiased estimates and improves the generalizability of results (Agency for Healthcare Research and Quality, 2020).

Recent advancements in healthcare analytics emphasize the role of data-driven models in identifying high-risk populations and supporting decision-making (Bates et al., 2014; Wang & Alexander, 2015).

This study builds on existing literature by combining survey-weighted modeling with temporal validation and risk stratification techniques.

RESEARCH METHODS

The study utilized longitudinal data from five MEPS panels. Data preprocessing included handling missing values, recoding categorical variables, and creating derived variables such as total two-year expenditure and log-transformed expenditure.

Persistent high-cost individuals were defined as those in the top 10% of total two-year healthcare spending. Survey weights were adjusted by pooling across panels to maintain representativeness.

A survey-weighted logistic regression model was implemented using the svyglm function to account for stratification and clustering in the data. Predictor variables included demographic characteristics, insurance status, chronic conditions, and prior healthcare expenditure.

To improve external validity, a panel-based temporal split was used, with panels 1–4 used for training and panel 5 used for testing. Model performance was evaluated using AUC, sensitivity, specificity, precision, accuracy, and F1 score.

Risk stratification was performed by dividing predicted probabilities into quintiles to assess the model's ability to segment populations.

ANALYSIS RESULTS AND DISCUSSION

The model demonstrated strong predictive performance, achieving an AUC of 0.934, indicating excellent discrimination ability.

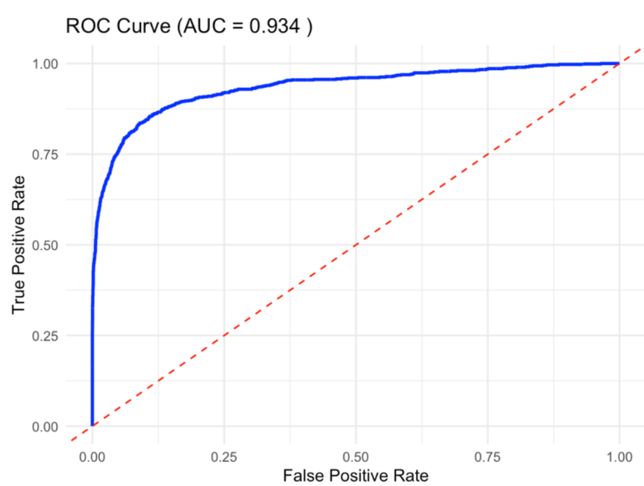

Figure 1: ROC Curve

At a threshold of 0.10, the model achieved a sensitivity of 90.1% and specificity of 80.8%. The high sensitivity ensures that most high-risk individuals are identified, which is critical in healthcare applications.

Although precision is lower due to class imbalance, this is expected and acceptable in risk prediction contexts where identifying high-risk individuals is the primary objective.

Risk stratification results show a strong gradient across groups.

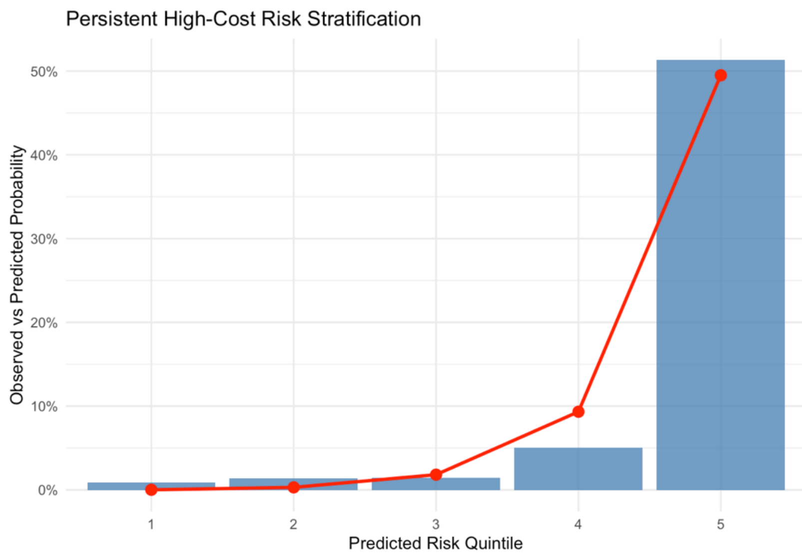

Figure 2 – Risk Stratification Plot

Individuals in the highest risk quintile had an observed high-cost rate of over 50%, compared to less than 1% in the lowest quintile. This demonstrates the model's strong ability to segment populations into actionable categories.

Predictor analysis revealed that prior healthcare expenditure was the strongest factor, with an odds ratio greater than 4.

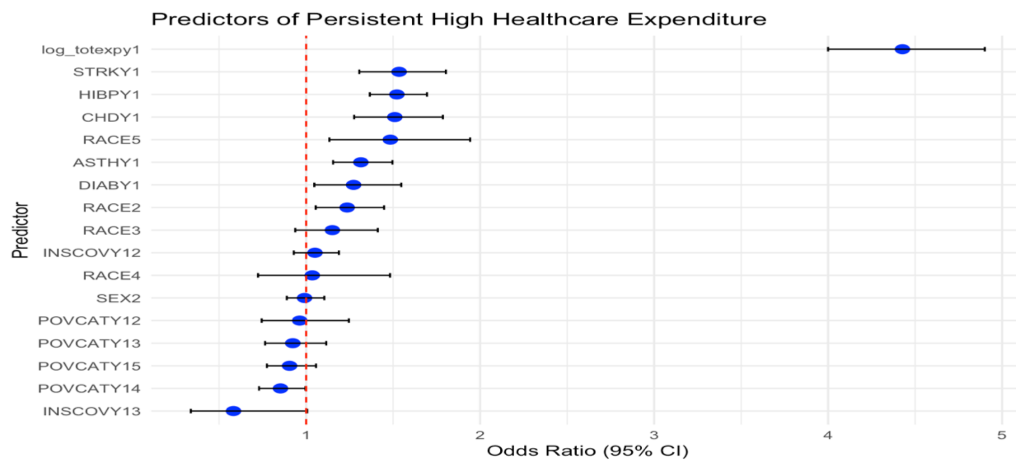

Figure 3 – Odds Ratio Plot

Chronic conditions such as stroke, heart disease, hypertension, diabetes, and asthma were also significant predictors, aligning with existing healthcare literature.

Panel-level analysis showed consistent patterns across time.

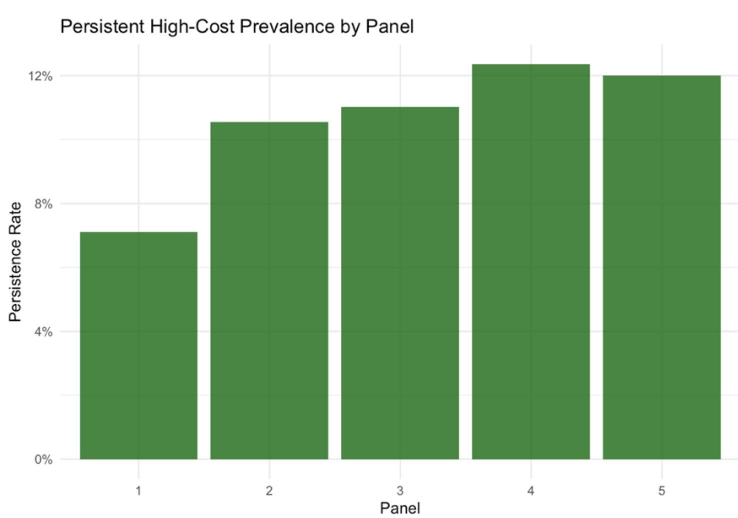

Figure 4 – Panel Prevalence Plot

Overall, the model provides both strong predictive accuracy and interpretability, making it suitable for healthcare applications.

DASHBOARD IMPLEMENTATION

The predictive model was operationalized through an interactive Shiny dashboard designed to support data-driven healthcare decision-making. The dashboard provides an integrated view of model performance, risk assessment, and actionable insights derived from the analysis. It highlights key performance indicators, including an AUC of 0.934, sensitivity of 90.1%, specificity of 80.8%, and overall accuracy of 81.9%, complemented by a ROC curve visualization to assess model discrimination. In addition, the dashboard presents a concise summary of the project objective and outcome definition, along with risk-based insights such as stratification of individuals into risk quintiles and identification of key predictors based on odds ratios. This integrated approach enhances interpretability by translating complex analytical results into intuitive visual insights, enabling effective identification and prioritization of individuals at high risk of persistent healthcare expenditure for targeted interventions and optimized resource allocation.

Live Dashboard Link: https://capstonealy6980.shinyapps.io/capstone-dashboard/

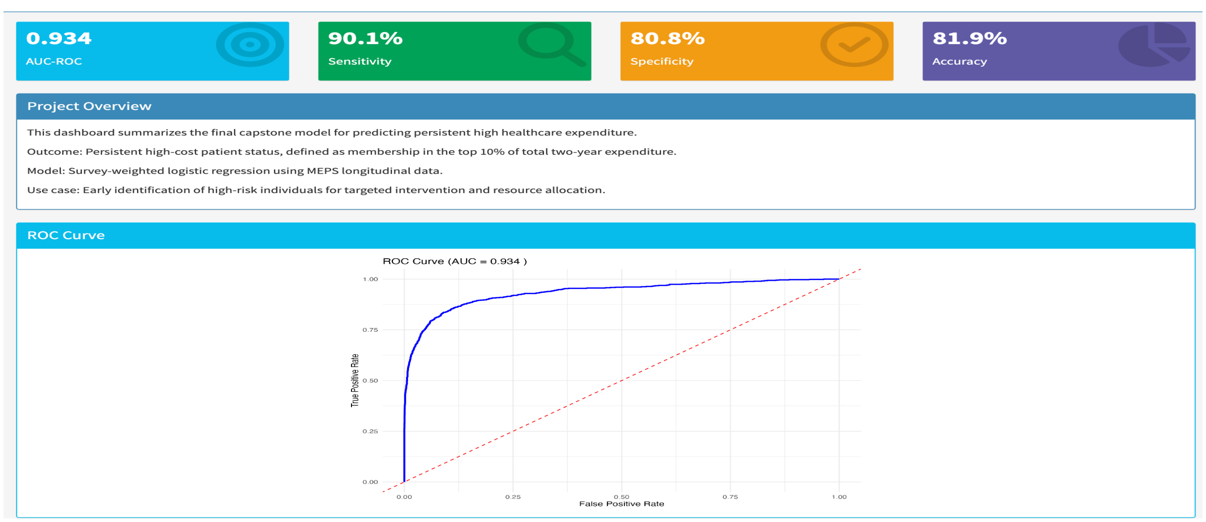

Figure 5: Dashboard Overview with Model Performance Metrics

The dashboard enables stakeholders to interpret model outputs efficiently and identify high-risk individuals for targeted interventions. By combining performance metrics with visual insights, the tool bridges the gap between predictive modeling and real-world healthcare applications. It can support decision-makers in prioritizing care management strategies, optimizing resource allocation, and improving overall healthcare outcomes.

CONCLUSION

This study developed an interpretable predictive model to identify individuals at risk of persistent high healthcare expenditure using longitudinal MEPS data. By applying survey-weighted logistic regression with a panel-based validation approach, the model achieved strong performance, with an AUC of 0.934, indicating excellent discrimination. High sensitivity at the selected threshold ensured that most high-risk individuals were correctly identified, while risk stratification demonstrated clear separation across population groups.

The findings highlight prior healthcare expenditure and chronic conditions such as hypertension, stroke, heart disease, diabetes, and asthma as key drivers of long-term costs. These insights can support targeted interventions, improved care management, and efficient resource allocation. Overall, the study demonstrates the practical value of predictive analytics in healthcare and provides a scalable framework for identifying high-risk populations and supporting data-driven decision-making.

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APPENDIX

APPENDIX A: Confusion Matrix

The confusion matrix summarizes the classification performance of the predictive model by comparing actual and predicted outcomes for persistent high-cost status. The model was evaluated at a threshold aligned with the outcome prevalence (10%), ensuring that high-risk individuals are effectively identified.

	Predicted High-Cost	**Predicted Non-High-Cost**
Actual High-Cost	True Positive (TP)	False Negative (FN)
Actual Non-High-Cost	False Positive (FP)	True Negative (TN)

The model achieved high sensitivity (90.1%), indicating that the majority of true high-cost individuals were correctly identified. While precision is lower due to class imbalance, this is expected in healthcare risk prediction where minimizing missed high-risk cases is prioritized.

APPENDIX B: Model Performance Metrics

Figure B1: Model Performance Metrics

Performance Metrics

	Metric	Value
1	AUC-ROC	0.9343
2	Sensitivity	0.9006
3	Specificity	0.8077
4	Precision	0.3899
5	Accuracy	0.8188
6	F1 Score	0.5442

The model demonstrates strong predictive performance with an AUC of 0.934, indicating excellent discrimination between persistent high-cost and non-high-cost individuals. The sensitivity of 90.1% ensures effective identification of high-risk patients, while specificity (80.8%) and accuracy (81.9%) indicate balanced classification performance. The F1 score reflects the trade-off between precision and recall under class imbalance.

APPENDIX C: Data Dictionary

The following variables were used in the predictive model derived from MEPS longitudinal data:

Variable Name	Description
TOTEXPY1	Total healthcare expenditure in Year 1
TOTEXPY2	Total healthcare expenditure in Year 2
log_totexpy1	Log-transformed Year 1 expenditure (key predictor)
AGEY1X	Age of the individual
SEX	Gender indicator
RACE	Race/ethnicity category
INSCOVY1 / INSCOVY2	Insurance coverage status
DIABY1	Diabetes indicator
HIBPY1	Hypertension indicator
CHDY1	Heart disease indicator
ASTHY1	Asthma indicator
STRKY1	Stroke indicator
Outcome Variable	Persistent high-cost status (top 10% of two-year total expenditure)

These variables were selected based on clinical relevance and prior literature, with a strong emphasis on prior healthcare expenditure and chronic conditions as key drivers of cost persistence.